## Step 1: Install PostgreSQL and pgAdmin (One-Time Setup)

If you already have PostgreSQL and pgAdmin installed on your laptop, skip straight to Step 2.

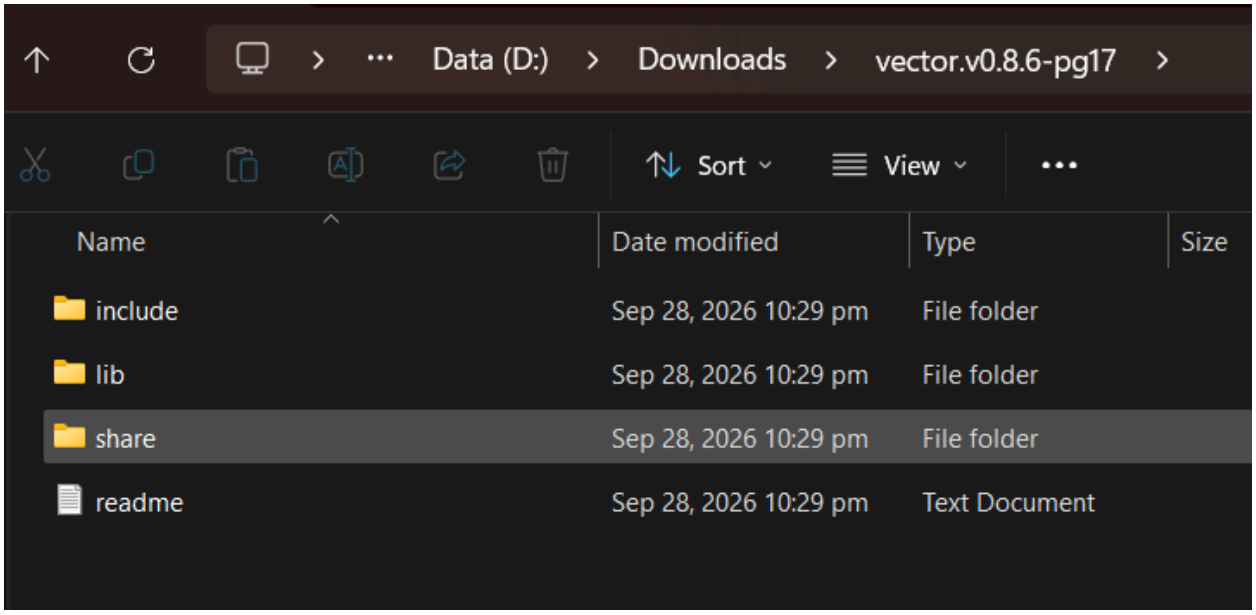
1. Go to the PostgreSQL Downloads page and select your OS (Windows or macOS). Choose Postgres v17.11.
2. Download and run the installer.
3. Keep the default options checked
    a. **PostgreSQL Server**,
    b. **pgAdmin 4**,
    c. **Command Line Tools**
4. When prompted for a password for the `postgres` user, enter an easy password like `postgres` or `admin123`. Keep the port as **5432**.
5. Finish the setup.

## Step 2: Install the pgvector Extension

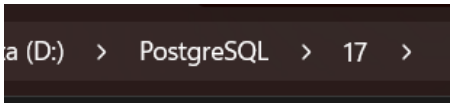
1. Download the precompiled package for PostgreSQL 17:
    - Go to github.com/andreiramani/pgvector_pgsql_windows/releases.
    - Download the ZIP under the release for pgvector for PostgreSQL 17.




2. Extract the downloaded ZIP. Inside, locate the matching folders:
    - vector.dll (inside the lib folder)
    - vector.control and all vector--*.sql files (inside the share/extension folder)



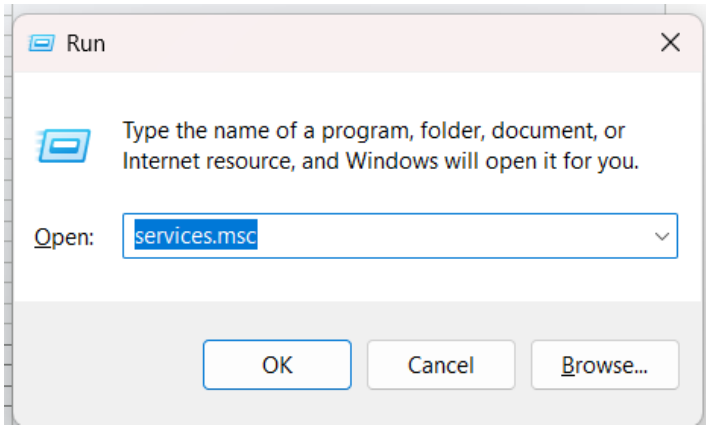

3. Go to your PostgreSQL 17 directory:



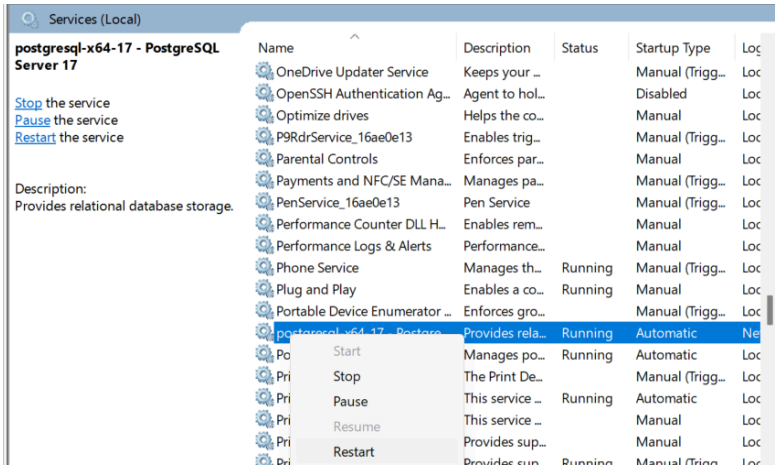

4. Copy and paste them into your actual PostgreSQL 17 directory
   - Copy vector.dll → paste into : **..\PostgreSQL\17\lib**

     *(If Windows asks for Administrator permission, click Continue)*

   - Copy vector.control and all vector--*.sql files → paste into: **..\PostgreSQL\17\share\extension**
5. Restart the PostgreSQL Service:
   - Press Win + R, type services.msc, and press Enter.

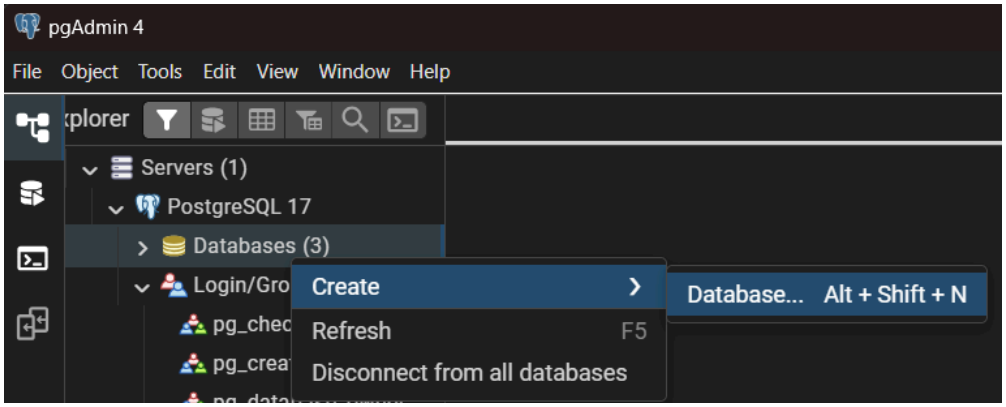


- Find postgresql-x64-17 in the list.
- Right-click it and click Restart.

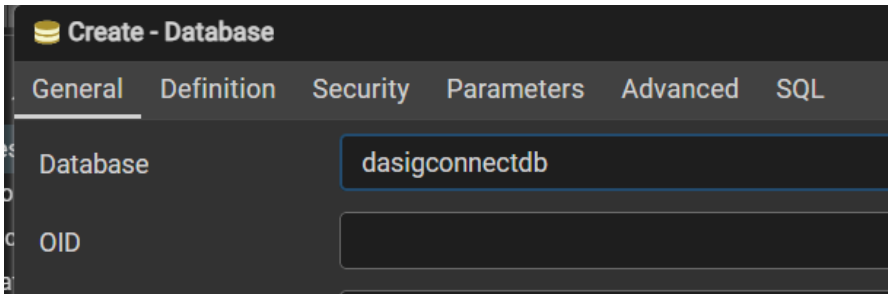



## Step 3: Create the Local Database in pgAdmin

1. Open **pgAdmin 4**.
2. Click **Servers** on the left menu and enter the master password you set during installation.
3. Right-click on **Databases** → **Create** → **Database…**



4. Name the database:




5. Click **Save**.

## Step 4: Update Your Local `.env` File

Open your local `.env` file inside the `backend` folder. Change the database variables so they point to your own machine instead of Supabase:

DASIG_DATABASE_URL=jdbc:postgresql://localhost:5432/**dasigconnectdb #your db name**
DASIG_DATABASE_USER=postgres
DASIG_DATABASE_PASSWORD=**YOUR_POSTGRES_PASSWORD**

*Replace YOUR_POSTGRES_PASSWORD with the password you created during the PostgreSQL setup in Step 1.*

## Step 5: Run the Backend

1. Update env first:

   #to create the seeded admin put this in the env or uncomment
   # LOCAL DEV: true = seed the protected institution + a bootstrap admin on startup.
   APP_DB_DIAGNOSTICS_ENABLED=true
   BOOTSTRAP_ADMIN_EMAIL=admin@dasigconnect.com
   BOOTSTRAP_ADMIN_PASSWORD=admin1234567  #12+ char password

2. Run the backend once:

   cd backend
   ./mvnw spring-boot:run

3. Change APP_DB_DIAGNOSTICS_ENABLED=true to **false** to stop seeding another admin.
4. Then restart the backend

**FOR SEEDING TEST USERS:**

1. Create backend/dev/users.sql
2. Copy and paste this to users.sql:

***DO NOT PUSH THIS TO REPO.***

```
-- Development-only user seed.
-- Run manually after Flyway has created the schema:
--   psql -h localhost -U postgres -d dasigconnectdb -f backend/dev/users.sql
--
-- Do not add this file to Flyway migrations or run it against production.
-- Passwords are converted to BCrypt hashes by PostgreSQL pgcrypto.
CREATE EXTENSION IF NOT EXISTS pgcrypto;
-- Keep the protected default institution separate from contributor test data.
INSERT INTO institutions (name, code, status, email_domain)
VALUES (
        'DASIG Central Visayas',
        'DASIG-CV',
        'active',
        'dasig.gov.ph'
    ) ON CONFLICT (code) DO NOTHING;
INSERT INTO institutions (name, code, status, email_domain)
VALUES (
        'Development Institute One',
        'DEV-INST-1',
        'active',
        'dev-one.example'
    ) ON CONFLICT (code) DO NOTHING;
INSERT INTO institutions (name, code, status, email_domain)
VALUES (
        'Development Institute Two',
        'DEV-INST-2',
        'active',
        'dev-two.example'
    ) ON CONFLICT (code) DO NOTHING;
-- Admin: admin@dasigconnect.com / Dasigconnect123
INSERT INTO users (
        email,
        password_hash,
        first_name,
        last_name,
        role,
        account_state,
        is_admin
    )
VALUES (
        'admin@dasigconnect.com',
        crypt('Dasigconnect123', gen_salt('bf')),
        'DASIG',
        'Admin',
```

```
        'admin',
        'active',
        TRUE
    ) ON CONFLICT (email) DO NOTHING;
-- Moderators: moderator@dasigconnect.com, moderator2@dasigconnect.com,
-- moderator3@dasigconnect.com / Moderator123!
INSERT INTO users (
        email,
        password_hash,
        first_name,
        last_name,
        role,
        account_state,
        is_admin
    )
VALUES (
        'moderator@dasigconnect.com',
        crypt('Moderator123!', gen_salt('bf')),
        'Development',
        'Moderator',
        'moderator',
        'active',
        FALSE
    ) ON CONFLICT (email) DO NOTHING;
INSERT INTO users (
        email,
        password_hash,
        first_name,
        last_name,
        role,
        account_state,
        is_admin
    )
VALUES (
        'moderator2@dasigconnect.com',
        crypt('Moderator123!', gen_salt('bf')),
        'Development',
        'Moderator Two',
        'moderator',
        'active',
        FALSE
    ),
    (
        'moderator3@dasigconnect.com',
        crypt('Moderator123!', gen_salt('bf')),
        'Development',
        'Moderator Three',
        'moderator',
        'active',
        FALSE
    ) ON CONFLICT (email) DO NOTHING;
```

```
-- Two contributors per institution. All use Contributor123!.
INSERT INTO users (
        email,
        password_hash,
        first_name,
        last_name,
        role,
        institution_id,
        account_state,
        is_admin
    )
SELECT account.email,
    crypt('Contributor123!', gen_salt('bf')),
    account.first_name,
    account.last_name,
    'contributor',
    institution.id,
    'active',
    FALSE
FROM (
        VALUES (
                'contributor@dasigconnect.com',
                'Development',
                'Contributor One',
                'DEV-INST-1'
            ),
            (
                'contributor2@dasigconnect.com',
                'Development',
                'Contributor Two',
                'DEV-INST-1'
            ),
            (
                'contributor3@dasigconnect.com',
                'Development',
                'Contributor Three',
                'DEV-INST-2'
            ),
            (
                'contributor4@dasigconnect.com',
                'Development',
                'Contributor Four',
                'DEV-INST-2'
            )
    ) AS account(email, first_name, last_name, institution_code)
    JOIN institutions institution ON institution.code = account.institution_code ON
CONFLICT (email) DO NOTHING;
-- Correct existing rows if this seed was previously run with DASIG-CV.
UPDATE users
SET institution_id = institution.id
FROM institutions institution
```

```
WHERE users.email IN (
        'contributor@dasigconnect.com',
        'contributor2@dasigconnect.com'
    )
    AND institution.code = 'DEV-INST-1';
UPDATE users
SET institution_id = institution.id
FROM institutions institution
WHERE users.email IN (
        'contributor3@dasigconnect.com',
        'contributor4@dasigconnect.com'
    )
    AND institution.code = 'DEV-INST-2';
SELECT email,
    role,
    account_state
FROM users
WHERE email IN (
        'admin@dasigconnect.com',
        'moderator@dasigconnect.com',
        'moderator2@dasigconnect.com',
        'moderator3@dasigconnect.com',
        'contributor@dasigconnect.com',
        'contributor2@dasigconnect.com',
        'contributor3@dasigconnect.com',
        'contributor4@dasigconnect.com'
    )
ORDER BY email;
```

3. To be safe.Open git ignore and add:

```
dev/
```

4. Run the file. Locate your Postgresql Folder. Open powershell.

```
 & "D:\PostgreSQL\17\bin\psql.exe" `            #should be your postgresql path
>>   -h localhost -U postgres -d dasigconnectdb `          #your db name
>>   -f .\backend\dev\users.sql
```

5. Enter your postgresql password from step 1.

```
PS D:\Documents\SCHOOL\3rdYrProjects\DASIGConnect> & "D:\PostgreSQL\17\bin\psql.exe" `
>>    -h localhost -U postgres -d dasigconnectdb `
>>    -f .\backend\dev\users.sql
Password for user postgres:

psql:backend/dev/users.sql:7: NOTICE:  extension "pgcrypto" already exists, skipping
CREATE EXTENSION
INSERT 0 0
INSERT 0 1
INSERT 0 1
INSERT 0 0
INSERT 0 1
INSERT 0 2
INSERT 0 4
UPDATE 2
UPDATE 2
            email             |    role     | account_state
------------------------------+-------------+---------------
 admin@dasigconnect.com       | admin       | active
 contributor@dasigconnect.com | contributor | active
 contributor2@dasigconnect.com | contributor | active
 contributor3@dasigconnect.com | contributor | active
 contributor4@dasigconnect.com | contributor | active
```